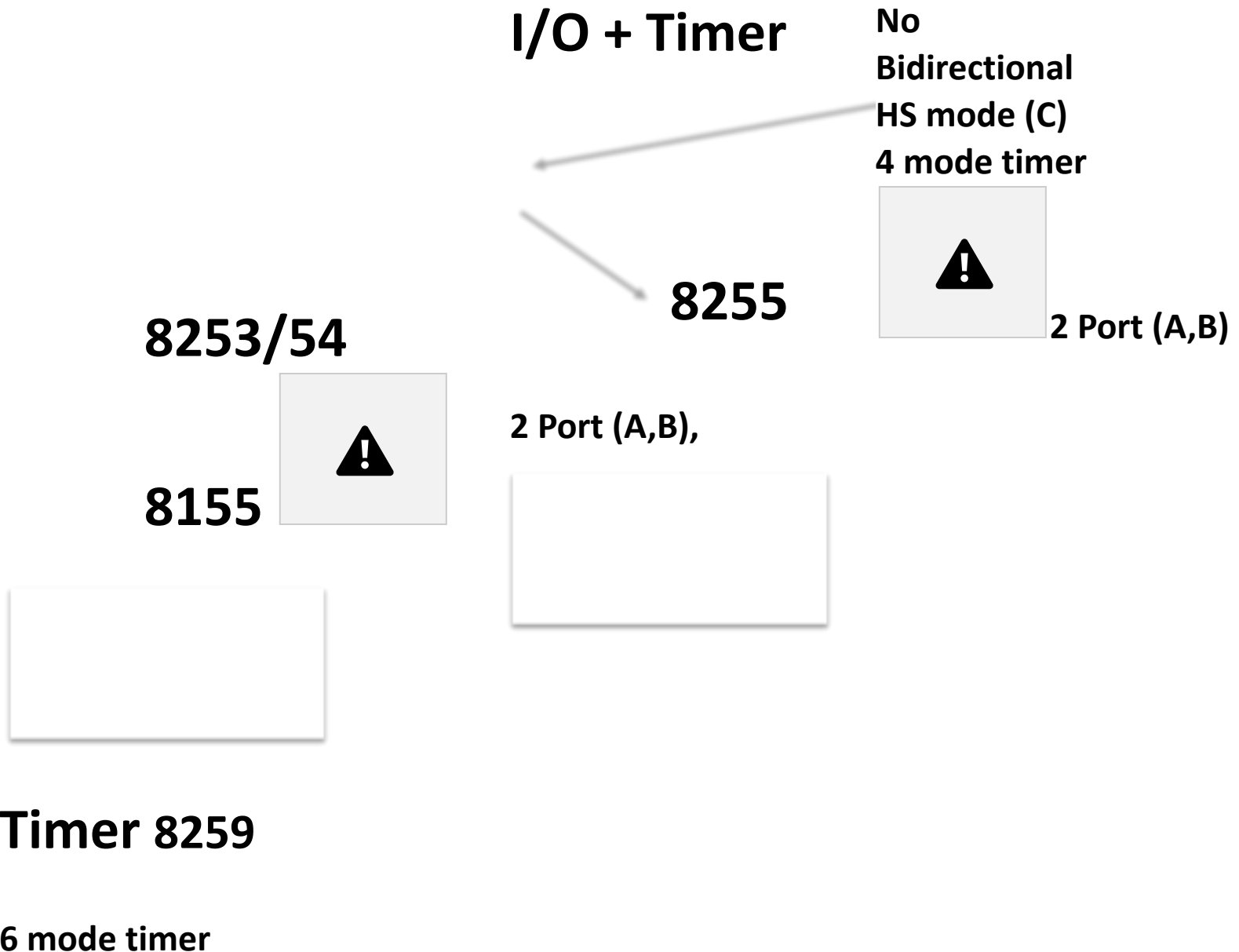


Hierarchy of I/O Control Devices





I/O



A is Bidirectional HS mode (C)

Extra controls **8251**



DMA controller

**Serial I/O USART
controller**

Interrupt controller

- **Parallel Vs Serial Communication**



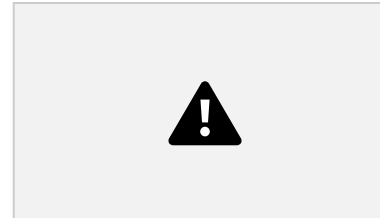
- **Characteristics of serial communication** – Synchronous/A-synchronous, Simplex/Duplex, Baud rate and Error Correction
- **Introduction to 8251 USART controller**



- Serial
 - Cheaper
 - Slower
- Parallel
 - Faster



Parallel



Serial

Data skew



Data
Transmission



—

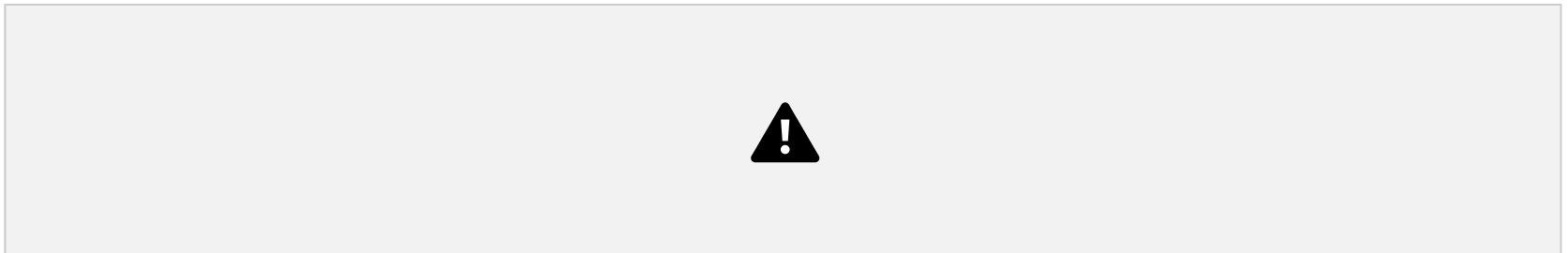
Limited to small distances



Synchronous



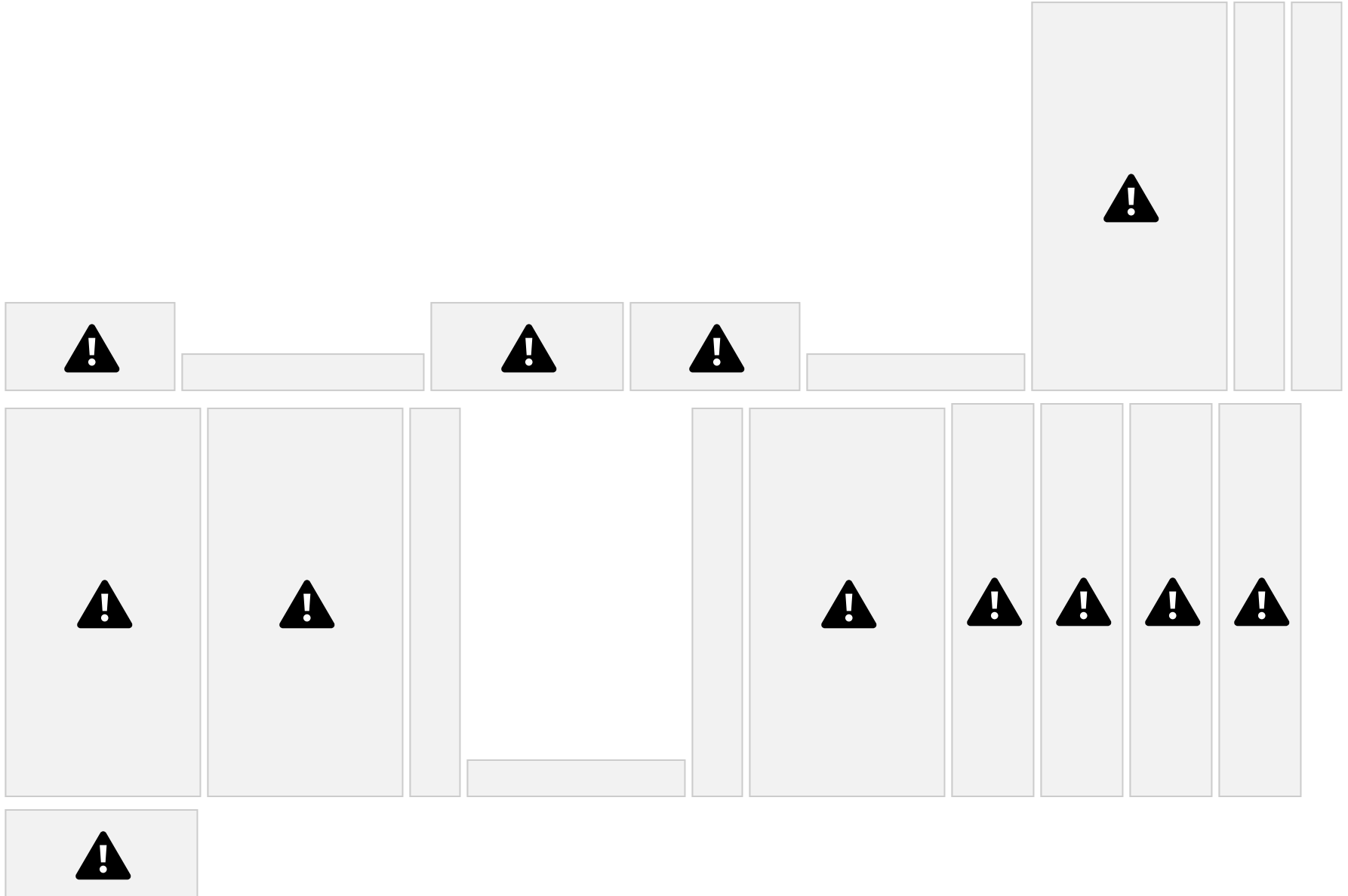
ASynchronous

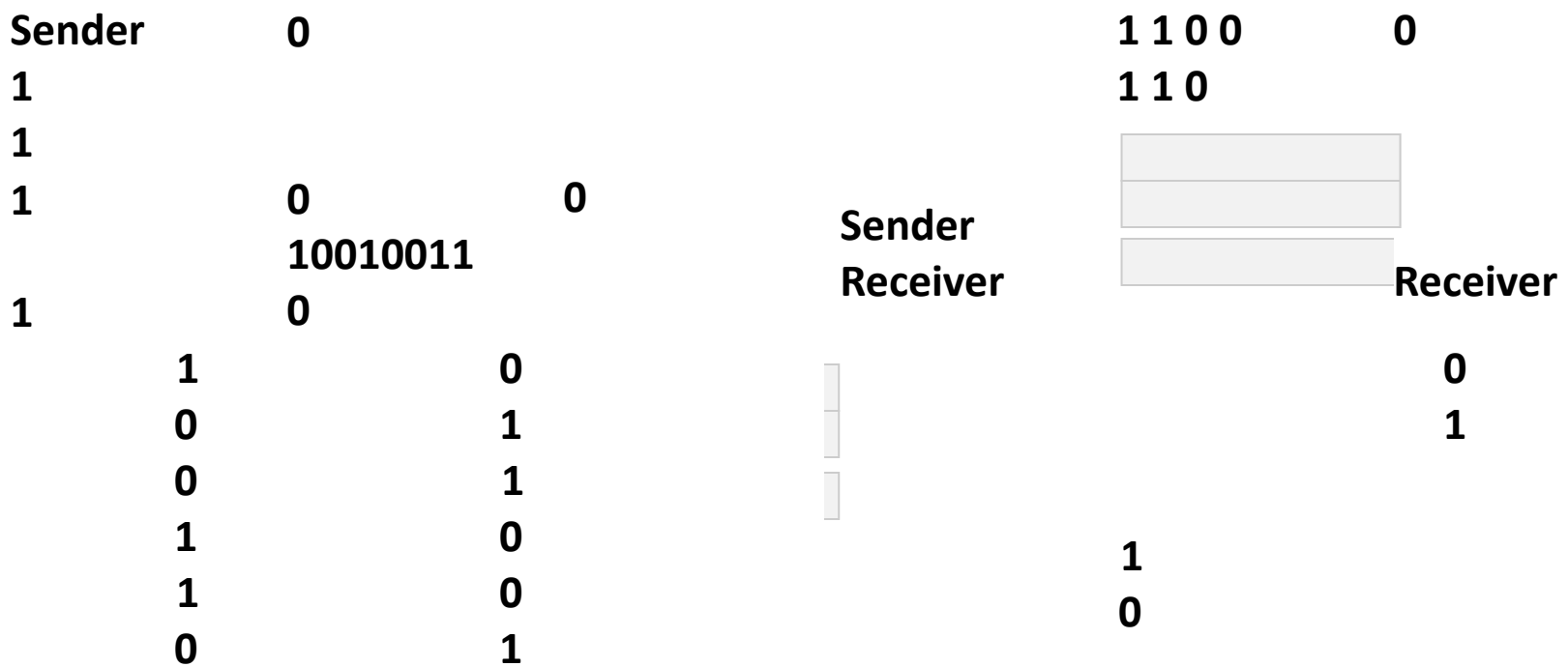


Two basic modes of data transmission

**Parallel to serial
Conversion**

**Serial to
parallel
Conversion**





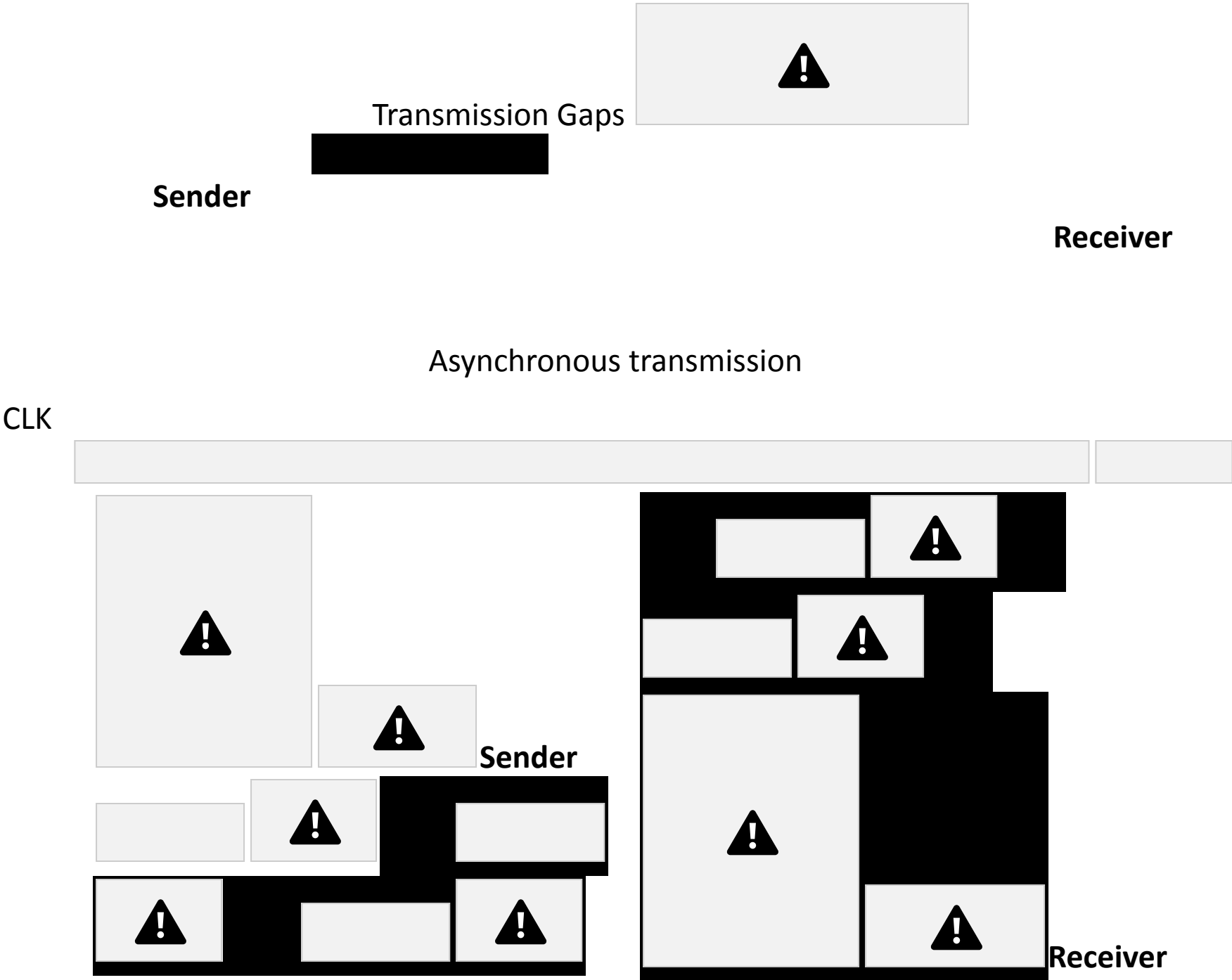
Serial Transmission Parallel Transmission



Synchronous – Sender and receiver must synchronize

- Done in hardware using phase locked loops (PLLs)
- Block of data can be sent
- More efficient : Less overhead than asynchronous transmission
- Expensive
- Asynchronous
 - Each byte is encoded for transmission
 - Start and stop bits
 - No need for sender and receiver synchronization







- Character oriented
 - Each character carried start bit and stop bits •
- When No data are being transmitted – Receiver stay at logic 1 called mark, logic 0 is Space • Framing:
- Transmission begins with one start bit (low/0)
 - Followed by DATA (8bit) and
 - Stop bits (1 or 2 bits of logic high)



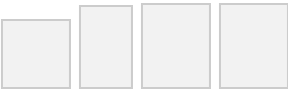
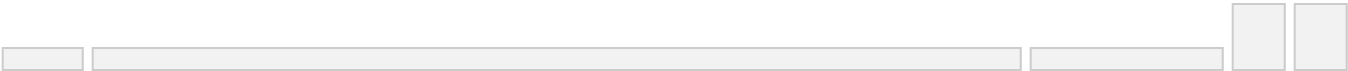
1 start

transmission



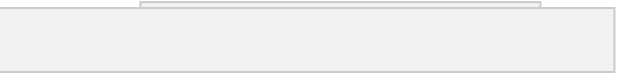
bit1 or 2 Stop Source data

bit

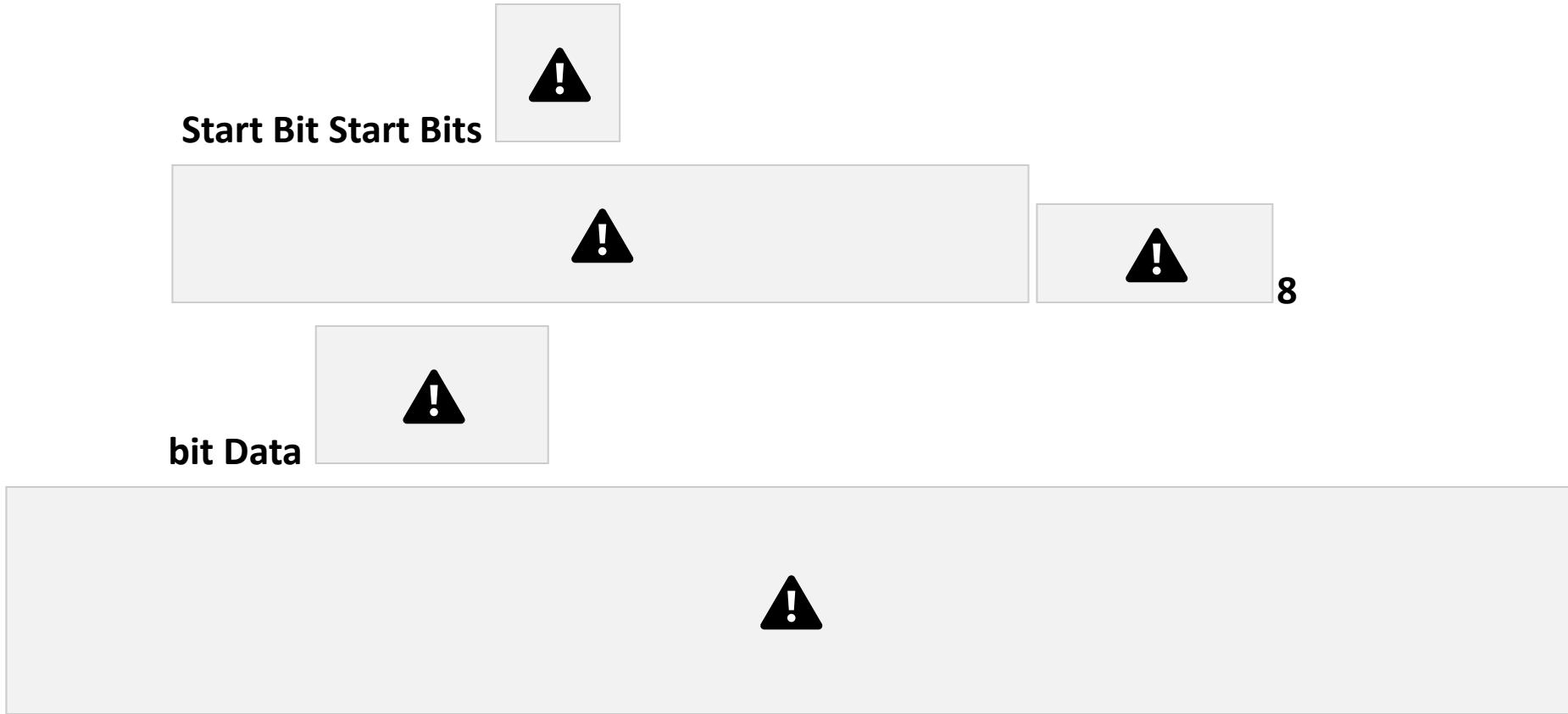


1 0 0 0 1 1 1 0

LSB MSB



Time



- Simplex
 - Data are transmitted in one directions
 - Example: CPU to printer
- Duplex
 - Data flow in both direction

- Half Duplex (Transmission goes on way at a time)
- Full Duplex (Both ways simultaneously)



•

Rate at which bits are transmitted (BAUD)

- Number of signal changes per second
- Bit time: how long the Bit stay On or Off
- Printer, Terminal Baud Adjustable (50-9600)
- 1200Baud means: Bit stay for $1/1200=0.83\text{ms}$



- Parity Check – Even parity: When odd numbers of 1 make $D7=1$ • Send Even number of 1
 - Odd parity: When even number of 1 make $D7=1$
 - Send Odd number of 1
- Check Sum
 - Used for block of data
 - Sum of all Bytes without carry and 2's complements
 - Total Sum Result should be Zero
- Cyclic Redundancy Code (CRC)
 - Synchronous Communication
 - Stream of Data can be represented by Cyclic polynomial that divided by a **constant polynomial** – Reminder to set **Bits** and Send out as check for error



- Inform RX the start bit, end bits and parity check
- Convert parallel word into stream of bits •
Create a transmit word by adding start, end and proper parity bit .
- Transmit one bit at a time with appropriate time delay using one data line
 - Time delay is determined by the speed of transmission

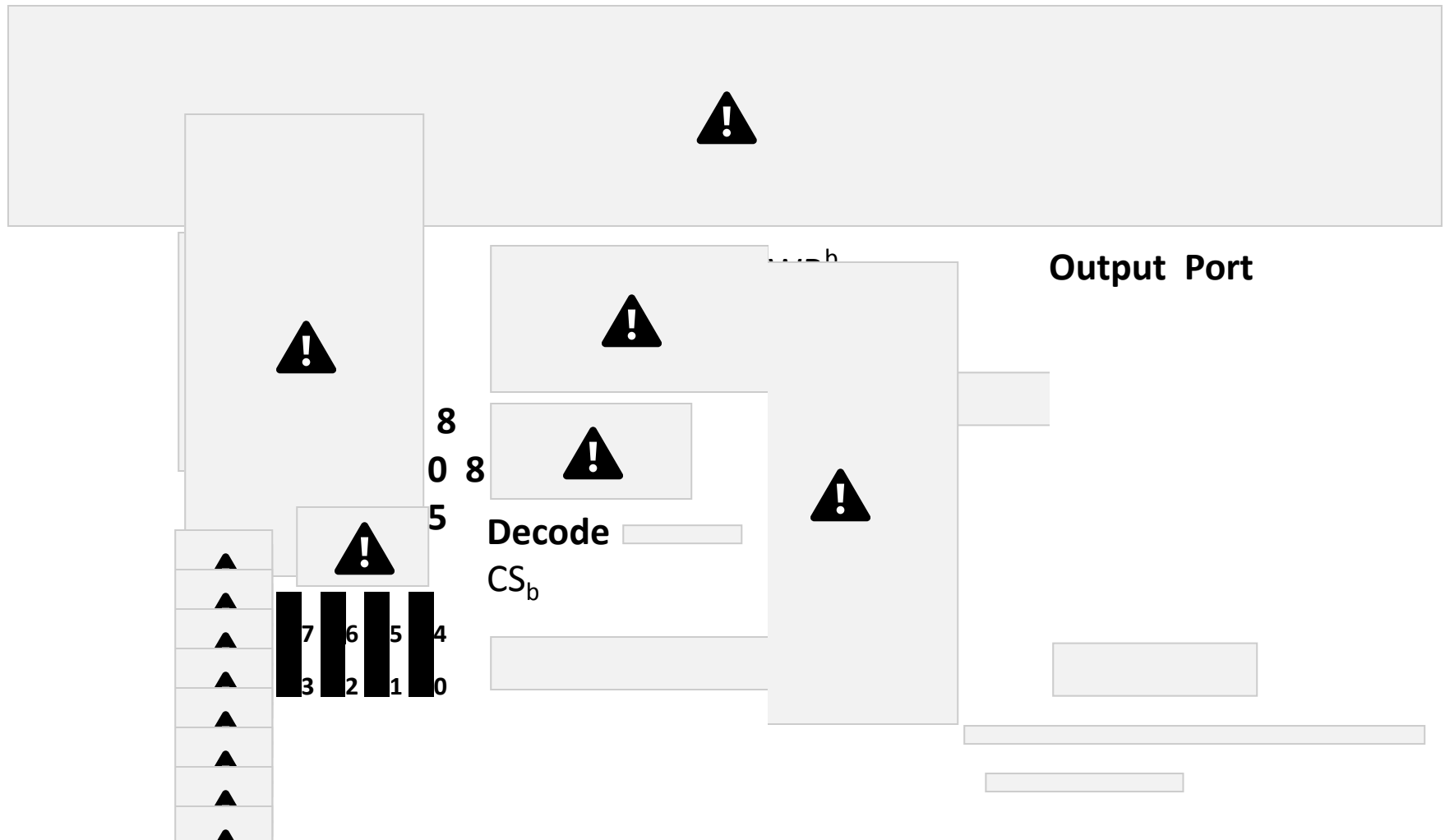


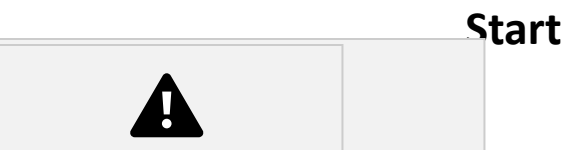
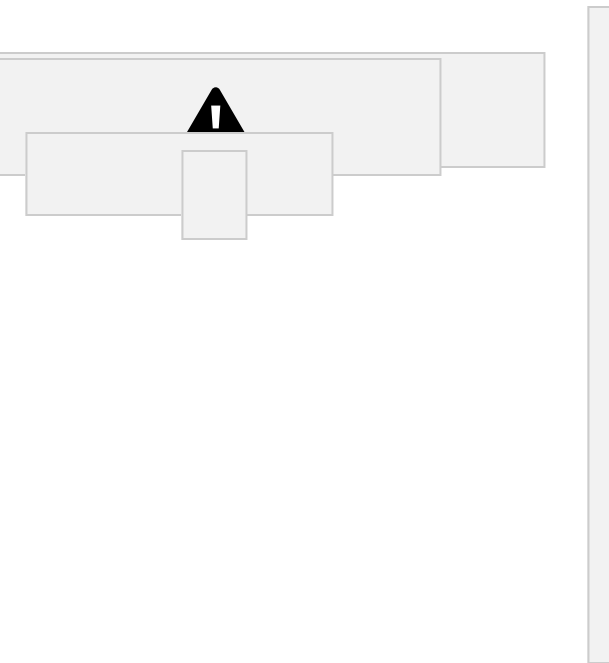
- Recognize bit of transmission
- Receive serial bits, one bit at a time
- Dismantle the start bits, end bit, parity bit, Data bits
- Check the error and recognize the end of transmission
- Convert serial data bit in to parallel word



- 8 bit Data to be send
- Steps:
 - Output a start bit

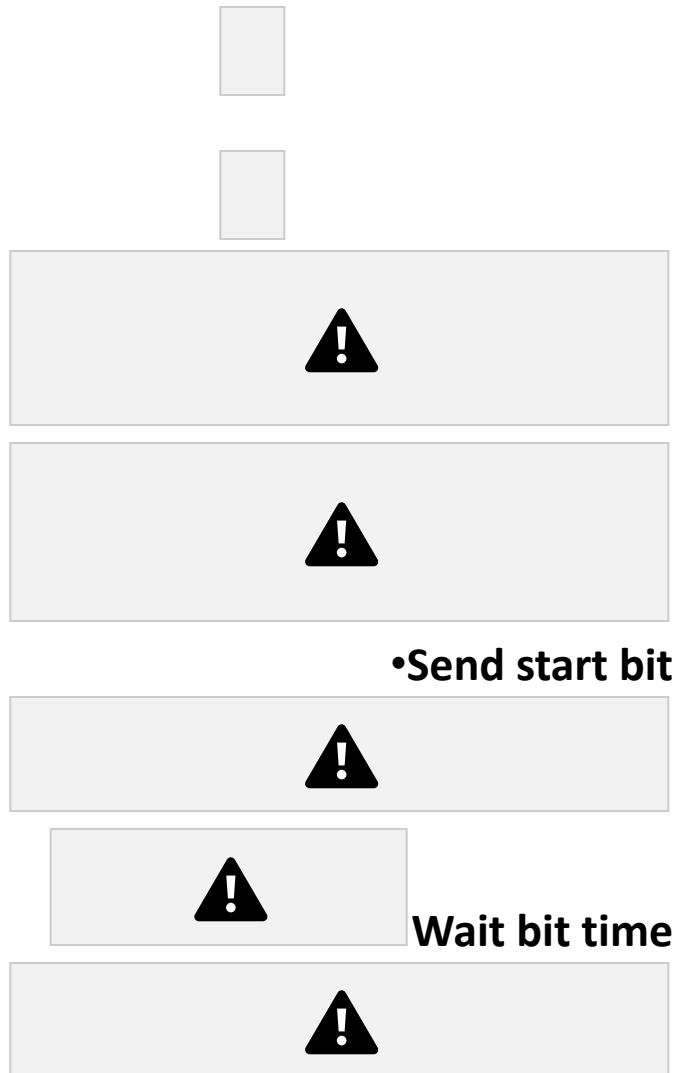
- Convert the character to be sent in a stream of serial bits with appropriate delay
- Add a parity information if needed
- Output one or two stop bit

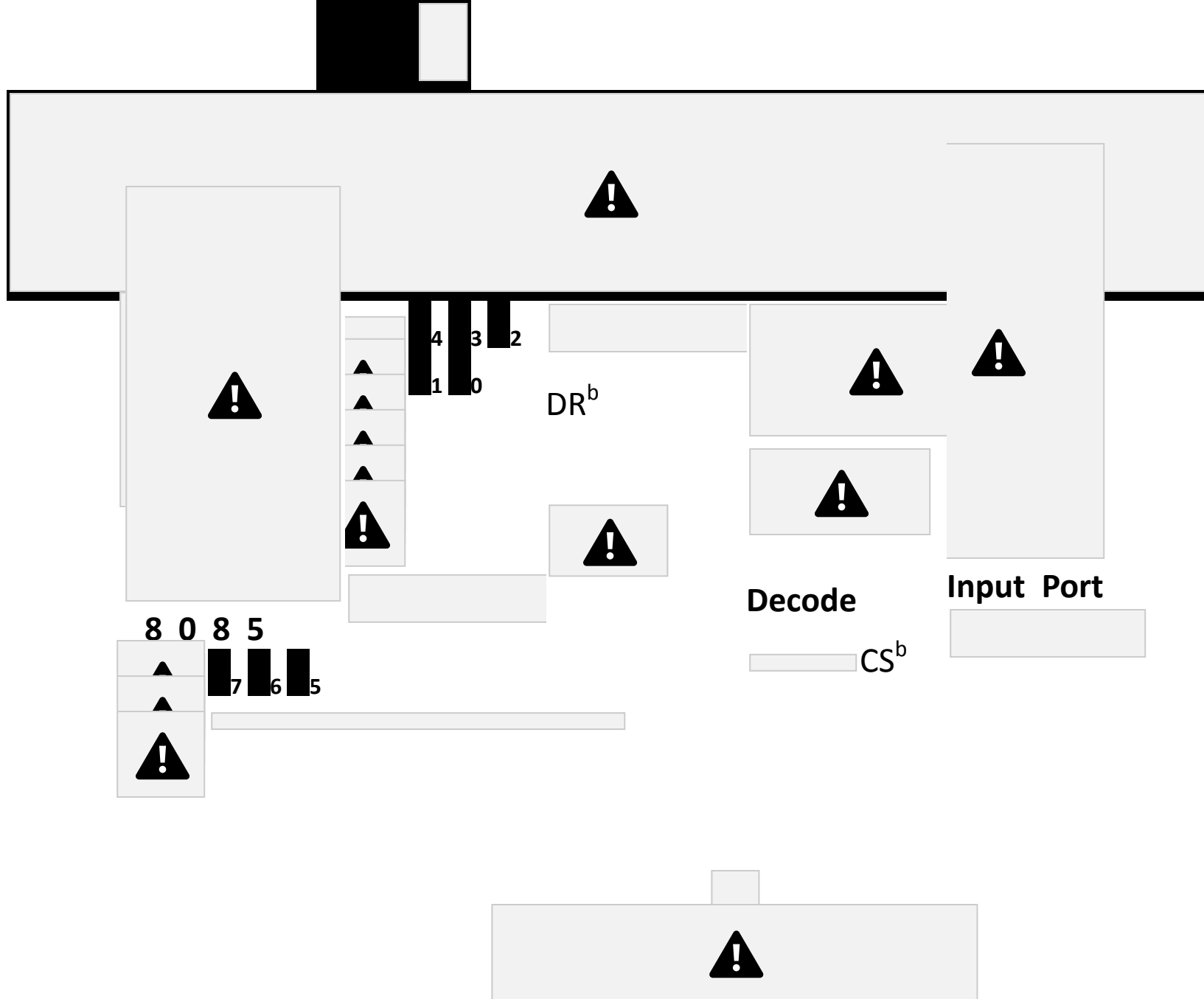


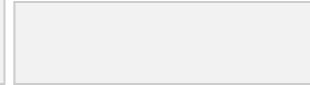
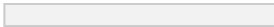
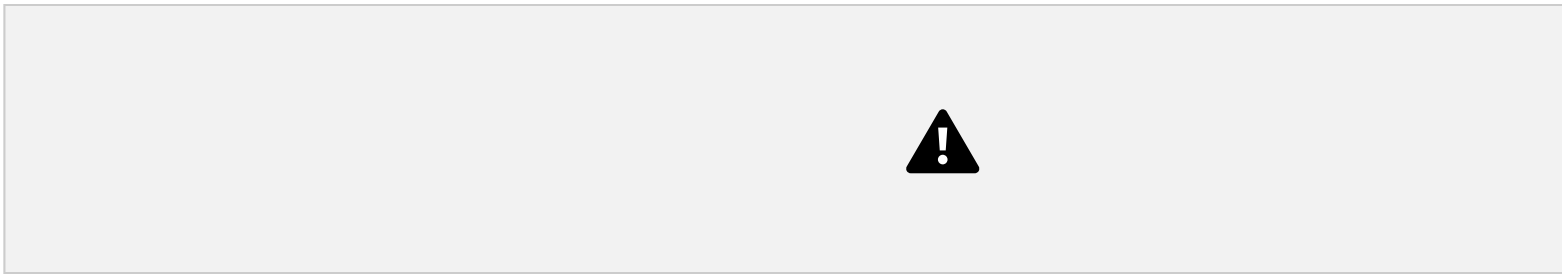


Wait bit time

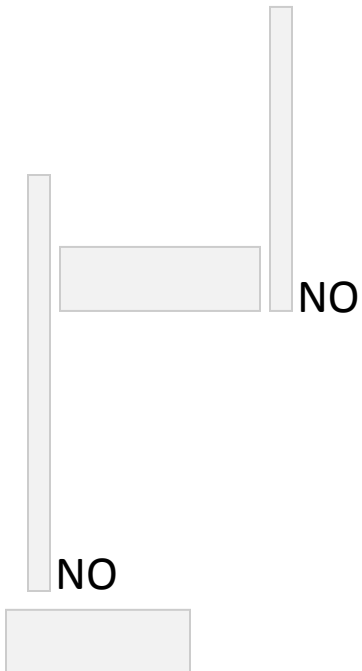
•Set up Character bit counter

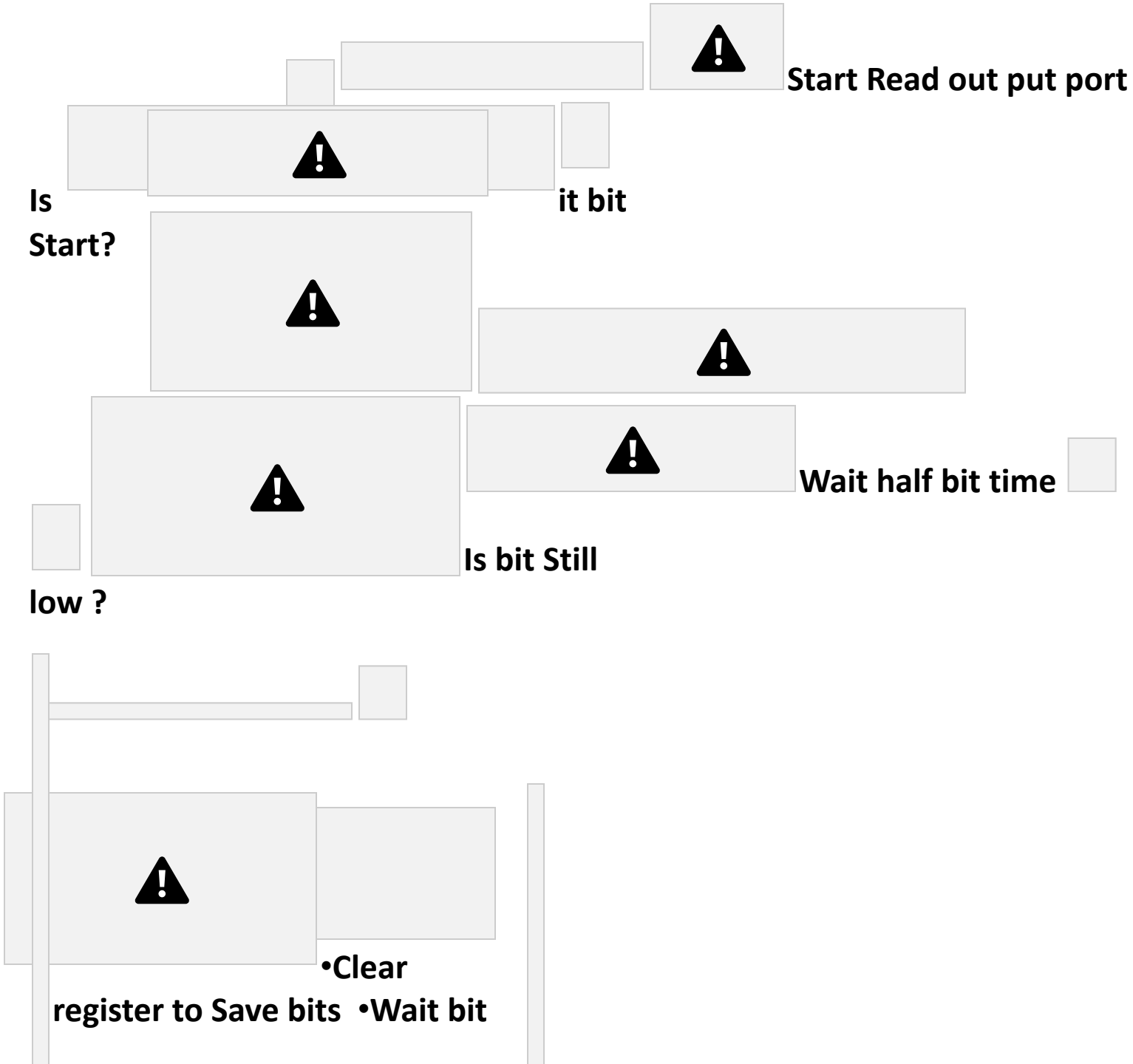


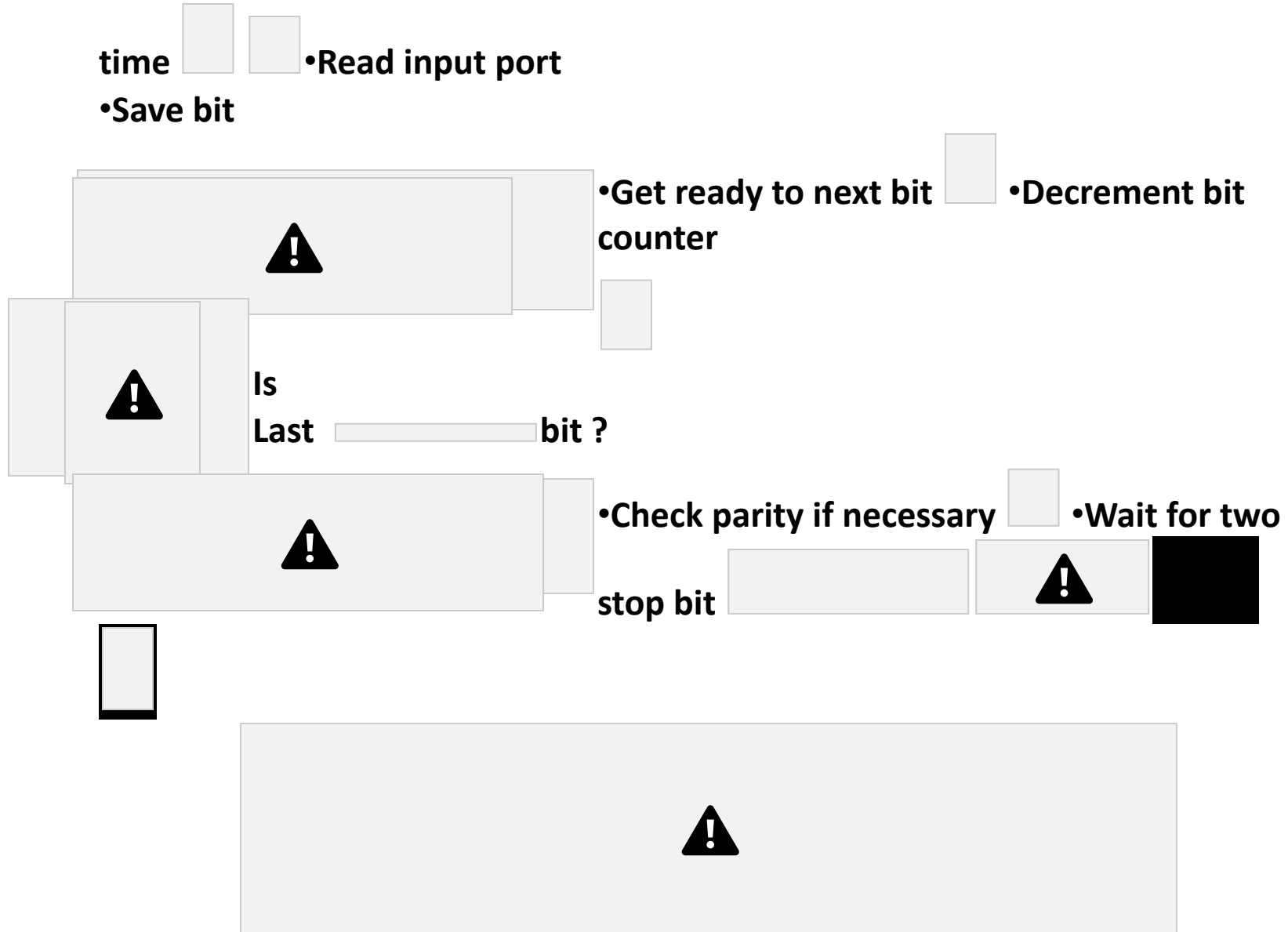




•Set up Bit counter







- Serial Input Data (SID)
- Serial Output Data (SOD)

- Instruction SIM is necessary to output data
- Interpretations (ACC contents)

SOD	SDE (0/1 Dis/Ena SOD)	X	For interrupts
-----	--------------------------	---	-------------------



MVI A, 80 ; Set D₇ in
the ACC=1
RAR ;Set D₆ =1
and bring carry

into D₇ SIM ; output D₇



Transmit an ASCII Char stored in Register B •



MVI B ASCIIDataByte ;
get data byte in B
MVI C,0BH ; set up
counter for 11 bits
XRA A ; reset carry to 0
NXTbit: MVI A,80H ;set
D7=1 in ACC
RAR
;bring
Carry
in D7
and
set
D6=1
SIM

;output D7
CALL DELAYBittime ;wait for fixed time (BWT)
STC ;set Carry 1
MOV A,B ;Place ASCII car in acc
RAR ; place ASCII D0 in Carry
;and shift 1 in D7
MOV B,A ;Save B

DCR C
JNZ NXTbit
RET

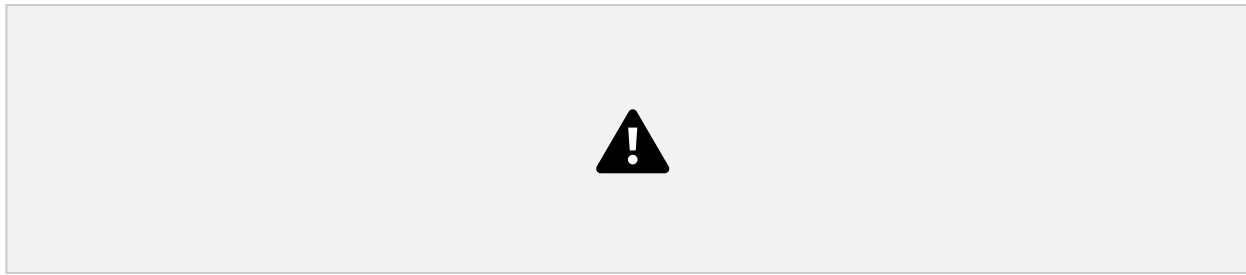


- Programmable chip 8251
- Requirement of HW control serial I/O – An input/output port are required for interfacing – Converts data bits in to Parallel to serial & vice versa
 - Data transfer to be synchronized between I/O
 - USART (Universal Synchronous Asynchronous Receiver and Transmitter)
- Writing a program compatible with all different serial



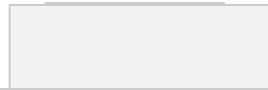
communication protocols is difficult and it is an inefficient use of microprocessor.

- UART: Universal Asynchronous Receiver/Transmitter chip.
- USART: Universal Synchronous/Asynchronous Receiver/Transmitter chip.
- The microprocessor sends/receives the data to the UART in parallel, while with I/O, the UART transmits/receive data serially.
- 8251 functions are integrated into standard PC interface chip.



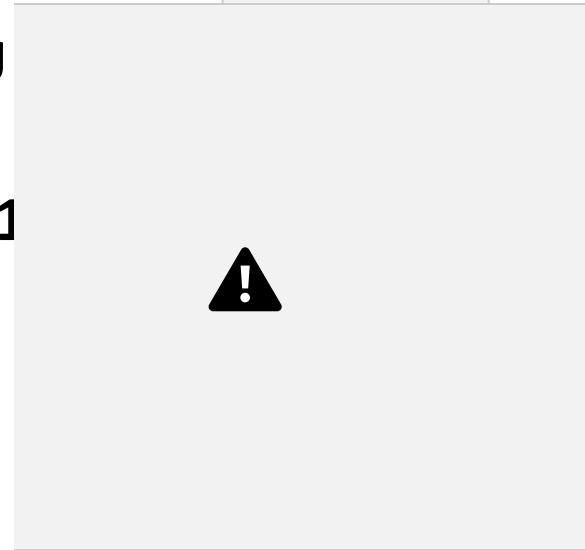
CPU

8251



**(8 bit)
serial port**

- UART/USART •
- 8251 USART



data

- 8250/16450 UART is a newer version of 8251.
- 16550 is the latest version UART.